AIST4010 Assignment 2: Antibiotic Resistance Gene (ARG) Classification

Experiment Report

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1 What the task was

15-class protein sequence classification from amino-acid FASTA: given a protein, predict whether it is an antibiotic resistance gene (ARG) and, if so, which of 14 antibiotic families it confers resistance to (aminoglycoside, MLS, polymyxin, fosfomycin, trimethoprim, bacitracin, quinolone, multidrug, chloramphenicol, tetracycline, rifampin, beta_lactam, sulfonamide, glycopeptide; label 14 = non-ARG). Train 21,209 sequences (10,067 ARG with drug annotation, 11,142 non-ARG), val 4,387 (same skewed mix), test 4,469. Metric is accuracy, and the test set is *class-balanced* ($\sim$298 per class). I ended with a two-model ensemble of class-balanced logistic-regression probes on ESM-2 650M and 3B embeddings: 98.10% accuracy on a balanced proxy of val (macro-F1 0.9753).

2 What I tried, in order

(See Table 1 at the end of the document.)

3 Things that broke and how I fixed them

3.1 Misreading the task as binary classification

The training headers split ARG proteins by drug family (`>id|FEATURES|src|drug|gene|mechanism|label`), and I initially read the task as a binary ARG-vs-non-ARG problem, training on the trailing 0/1. The first submission scored 0.94%. The sample_submission file shows 15 distinct labels (0–14). I then found the same assignment from Spring 2024 and another student's 2026 repo: the label space is 14 drug families plus non-ARG, 15 classes in total, and this year's mapping matches the 2024 one. Lesson: before modeling, check the label vocabulary of the submission template—it defines the problem, not the training-file format.

3.2 Validating on a skewed val fooled me twice

Train and val share the same skewed mix (52% non-ARG), but the test set is balanced. Every model that learned the training prior scored great on val (98.8–99.1%) but produced heavily non-ARG-skewed test predictions. On a balanced test that caps accuracy at $\sum_c \min(p_c, 298)/4469 \approx 38\%$ — my second submission regressed. Two fixes: (i) train with *inverse-frequency class weights* so the model learns a uniform prior that matches the test; (ii) evaluate on a *balanced proxy* of val (uniform 200 per class) and select models/weights by macro-F1 there, never by raw val accuracy.

3.3 Fine-tuning hurt instead of helping

LoRA fine-tuning of ESM-2 650M with a class-balanced cross-entropy reached 98.43% raw val accuracy (macro-F1 0.954), and ensembling it with the probe models pushed raw val accuracy to 99.13% — a number that looks great but is distribution-mismatched. On the balanced proxy, the fine-tuned model (97.47%) was *worse* than the plain balanced probe (98.10%), and adding it to the ensemble never helped. The fine-tuner overfits the skewed training distribution; freezing ESM-2 and fitting a linear probe regularizes better at this data scale.

3.4 Small bugs

- `liblinear` doesn't support multiclass; the first 15-class probe crashed, switched to `lbfgs`.
- 3B checkpoint lacks tokenizer files (huggingface layout) — copied vocab/special tokens from the 650M checkpoint.
- hf-mirror rate-limits (429) and huggingface.co direct TLS resets; eventually downloaded the 3B shards with plain `curl -C -` resume.

4 What worked and what didn't

4.1 Worked (roughly in order of contribution)

1. ESM-2 650M mean-pool embeddings + LR (97.8% balanced-proxy, the single biggest jump over the k-mer baseline's 94.7%);
2. class-balanced training (fixes the train/test prior mismatch, the difference between 38%-capped and 98%);
3. balanced-proxy evaluation (the model selector that prevents choosing skewed-prior models);
4. ensembling the 650M and 3B probes 1:1 (+0.3 on balanced proxy via error decorrelation);
5. 3-mer TF-IDF + LR as a cheap diversity member and sanity baseline.

4.2 Didn't work

1. fine-tuning ESM-2 (LoRA) as a member of the ensemble (worse than the probe, no ensemble gain);
2. ESM-2 3B probe alone (no gain over 650M; only helps in the ensemble);
3. 4-gram features (no gain over 3-gram);
4. prior correction (multiplying class marginals to uniform) as a post-hoc fix instead of class-weighted training.

4.3 What I took away from it

- The submission template's label distribution is part of the problem specification: it reveals the test prior, and a model that ignores it can look great locally and fail on the leaderboard.

- For skewed-train/balanced-test setups, pick models by macro-F1 on a balanced proxy of the validation set — raw accuracy on the skewed val is a biased selector.
- Frozen pretrained embeddings + linear probe beat end-to-end fine-tuning when the label distribution is skewed and the task is near-linearly separable.
- "Strong and different" still applies: the two probes share the same architecture but different ESM-2 scales, and their errors decorrelate enough to pay off.

#	Method	Balanced-proxy acc	Worked?	Notes
1	3-mer TF-IDF + LR, 15 classes	94.73%	baseline	fast sanity check; non-ARG vs multidrug confusion dominates
2	ESM-2 650M embeddings + LR	97.82%	yes	frozen embeddings are already near-linearly separable
3	ESM-2 650M + LoRA fine-tune (class-weighted)	97.47%	no	overfits skewed train prior; worse than the probe
4	4-gram features added	–	no	no gain over 3-gram
5	ESM-2 3B embeddings + LR	98.17%	slightly	∼equal to 650M alone; gains only in the ensemble
6	650M + 3B probe ensemble (1:1)	98.10%	yes	error decorrelation between model scales
7	probe + fine-tune + k-mer ensemble	97.8–98.0%	no	fine-tuner drags the ensemble down
8	class-balanced training (inverse-freq weights)	–	yes	the fix for the skewed-prior problem
9	balanced-proxy model selection	–	yes	the selector that exposed the val illusion

Table 1: Experiment log. Balanced-proxy acc: accuracy on a uniform 200-per-class subsample of val (the best available proxy for the balanced test set).